Acceleration

Exam code: 0625 & 0972

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Acceleration

Extender tier only

- **Acceleration** describes how the **velocity** of an object changes over **time**
- Acceleration is defined as:

The rate of change of velocity

- In other words, acceleration is the change in velocity per unit time
- The acceleration of an object is often changing throughout an object's journey
- Therefore, is it often useful to know the **average acceleration**

$$a = \frac{\Delta v}{\Delta t}$$

- Where:
  - a = acceleration in metres per second squared (m/s^2)
  - Δv = change in velocity in metres per second (m/s)
  - Δt = time taken in seconds (s)

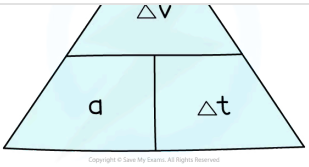
Formula triangle for acceleration, change in velocity and change in time

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



To use a formula triangle, simply cover up the quantity you wish calculate and the structure of the equation is revealed

- Information on how to use a formula triangle can be found in Speed & velocity

Change in velocity

- The **change in velocity** is the **difference** between the initial and final velocity:

$$\Delta v = v - u$$

- Where:
  - Δv = change in velocity in metres per second (m/s)
  - v = final velocity in metres per second (m/s)
  - u = initial velocity in metres per second (m/s)

Speeding up & slowing down

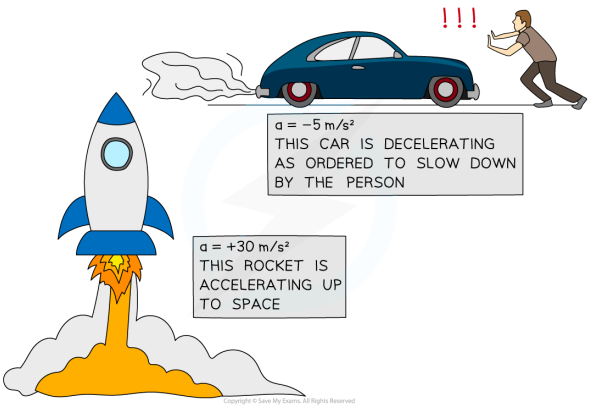
- An object can change its velocity in several ways:
  - speeding up
  - slowing down
  - changing direction

- Any change in an object's velocity is an **acceleration**
- When an object speeds up, it is **accelerating**
  - This is **positive acceleration**

- When an object slows down, it is **decelerating**
  - This is **negative acceleration**

- Acceleration is positive if its direction is in the **same direction** as the motion of the object

Acceleration of different objects



A rocket speeding up (accelerating) and a car slowing down (decelerating)

Worked Example

A Japanese bullet train decelerates at a constant rate in a straight line. The velocity of the train decreases from 50 m/s to 42 m/s in 30 seconds.

(a) Calculate the change in velocity of the train.

(b) Calculate the deceleration of the train, and explain how your answer shows the train is slowing down.

Answer:

Part (a)

Step 1: List the known quantities

- Initial velocity, $u = 50\ \mathrm{m/s}$
- Final velocity, $v = 42\ \mathrm{m/s}$

Step 2: Write the equation for change in velocity

$$\Delta v = v - u$$

Step 3: Substitute values for final and initial velocity

$$\Delta v = 42 - 50$$

$$\Delta v = -8\ \mathrm{m/s}$$

Part (b)

Step 1: List the known quantities

- Change in velocity, $\Delta v = -8\ \mathrm{m/s}$
- Time taken, $\Delta t = 30\ \mathrm{s}$

Step 2: Write the equation for acceleration

$$a = \frac{\Delta v}{\Delta t}$$

Step 3: Substitute the values for change in velocity and time

$$a = \frac{-8}{30}$$

$$a = -0.27\ \mathrm{m/s^2}$$

Step 4: Interpret the value for deceleration

- The answer is **negative**, which indicates the train is **slowing down**

Examiner Tips and Tricks

Remember, the units for acceleration are **metres per second squared**, $\mathrm{m/s^2}$. In other words, acceleration measures how much the velocity (m/s) changes every second, so the units are metres per second per second (m/s/s).

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